Day 6 — Java and .NET Agents, Custom Correlation

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Blueprint 1.6, 3.1, 3.2, 3.9, 3.10

Goal: install, configure, verify and upgrade the Java and .NET agents, and know when and how to use custom correlation.
Verification: every fact checked against the current Splunk AppDynamics docs (November 2026).

Configuration precedence (highest first)

1. **Environment variables**
2. **System properties** (-D... on the JVM command line)
3. Versioned agent.properties — <agent_home>/<ver>/conf/agent.properties
4. Global agent.properties — <agent_home>/conf/agent.properties
5. Versioned controller-info.xml — <agent_home>/<ver>/conf/controller-info.xml
6. Global controller-info.xml — <agent_home>/conf/controller-info.xml

The agent applies the **first non-empty value** it finds. Environment variables beat system properties, which beat the files.

Block 1 — Naming rules

- Application → Tier → Node. The agent tells the Controller which ones it belongs to.
- A node name must be **unique within the business application and the physical host**.
- Containers / short-lived JVMs: `-Dappdynamics.agent.reuse.nodeName=true` (optionally `...reuse.nodeName.prefix`) so new instances reuse node names.

Block 2 — Install the Java Agent (3.1)

1. **Unzip** to a folder the app user can write to (e.g. /opt/appdynamics/javaagent). Each release has its own ver<version>/ folder. Lock down: `chmod -R 700 <agent_home>/<ver>/conf`.
2. **Configure** <agent_home>/<ver>/conf/controller-info.xml (below).
3. **Load** the agent with -javaagent in the JVM start-up.
4. **Restart the JVM** — the agent only loads at JVM start.

```
<controller-host>ctrl.bank.local</controller-host>
<controller-port>8181</controller-port>        <!-- on-prem 8090/8181; SaaS 443 (or 80) -->
<controller-ssl-enabled>true</controller-ssl-enabled>
<application-name>PetClinic</application-name>
<tier-name>Web</tier-name>
<node-name>web01</node-name>
<account-name>customer1</account-name>         <!-- required for SaaS / multi-tenant -->
<account-access-key>xxxx</account-access-key>  <!-- required for ALL Controllers -->
```

Tomcat (Linux) — $CATALINA_BASE/bin/setenv.sh:

```
export CATALINA_OPTS="$CATALINA_OPTS -javaagent:/opt/appdynamics/javaagent/javaagent.jar"
```

- Windows: setenv.bat with `set CATALINA_OPTS=%CATALINA_OPTS% -javaagent:C:\appagent\javaagent.jar`.

- Tomcat as a Windows service: tomcat<ver>w.exe → Java tab → add the -javaagent line.

Block 3 — Three ways to set the same value

Setting	controller-info.xml	System property	Environment variable
Controller host	controller-host	-Dappdynamics.controller.hostName	APPDYNAMICS_CONTROLLER_HOST_NAME
Port	controller-port	-Dappdynamics.controller.port	APPDYNAMICS_CONTROLLER_PORT
SSL	controller-ssl-enabled	-Dappdynamics.controller.ssl.enabled	APPDYNAMICS_CONTROLLER_SSL_ENABLED
Account	account-name	-Dappdynamics.agent.accountName	APPDYNAMICS_AGENT_ACCOUNT_NAME
Access key	account-access-key	-Dappdynamics.agent.accountAccessKey	APPDYNAMICS_AGENT_ACCOUNT_ACCESS_KEY
Application	application-name	-Dappdynamics.agent.applicationName	APPDYNAMICS_AGENT_APPLICATION_NAME
Tier	tier-name	-Dappdynamics.agent.tierName	APPDYNAMICS_AGENT_TIER_NAME
Node	node-name	-Dappdynamics.agent.nodeName	APPDYNAMICS_AGENT_NODE_NAME

Use the XML for stable settings, system properties when several JVMs **share one agent folder** (each needs its own node name), and environment variables for **containers**.

Block 4 — Verify and troubleshoot

- Success: the node appears under **Tiers & Nodes** and starts reporting business transactions.
- Logs: `<agent_home>/ver<version>/logs/<node_name>/agent.log` — always read this first.

Symptom	Usual cause
Node never appears	Wrong host/port, firewall, wrong access key or account name
SSL errors in agent.log	controller-ssl-enabled vs port mismatch, or Controller certificate not trusted by the agent truststore (cacerts.jks)
Node under wrong application/tier	A higher-priority source (env var or -D) overrides controller-info.xml
Agent not loaded at all	-javaagent missing or wrong path, or JVM not restarted

Block 5 — Upgrade the Java Agent (3.9)

1. **Upgrade the Controller first**; upgrade agents on the tiers where BTs originate last.
2. Stop the app server; **back up** the agent folder.
3. Unzip the new agent to a new folder. **Rename** the old folder, then give the new folder the **old name** — the -javaagent path stays the same.
4. Copy controller-info.xml (and agent.properties if used). For app-agent-config.xml and other files, **copy your changes** into the new file — don't replace it.
5. SSL? Copy cacerts.jks too and make sure the agent user can read it.
6. **Restart the app server.** No Controller action needed.
7. After confirming, archive the old folder.

Block 6 — Install the .NET Agent on Windows (3.2)

- Installer **dotNetAgentSetup64.msi**, from an **elevated** prompt (Administrators group alone isn't enough). Install once per machine.
- **AppDynamics Agent Configuration utility:** Controller connection + **IIS** apps to instrument.
- Config: `%ProgramData%\AppDynamics\DotNetAgent\Config\config.xml` (edit as administrator). Logs: `%ProgramData%\AppDynamics\DotNetAgent\Logs`.
- **AppDynamics.Agent.Coordinator** service manages instrumentation. After a change: restart the coordinator, then the apps (iisreset for IIS).
- Windows services and standalone apps are **not** set up by the utility — add them to config.xml by hand:

```
<app-agents>
  <standalone-applications>
    <standalone-application executable="OrderProcessor.exe">
      <tier name="OrderProcessor"/>
    </standalone-application>
  </standalone-applications>
</app-agents>
```

Use the command-line attribute to tell apart two instances of the same executable; configure a **POCO entry point** if no framework is auto-detected.

Silent install:

```
msiexec /i dotNetAgentSetup64.msi /q /norestart /lv %TEMP%\AgentInstaller.log ^
        AD_SetupFile=C:\appd\setup.xml
net start AppDynamics.Agent.Coordinator
iisreset
```

Setup file root element: <winston>. Optional properties: INSTALLDIR, DOTNETAGENTFOLDER.

Block 7 — Upgrade the .NET Agent (3.10)

1. **From 3.9 or later:** no uninstall needed. **Before 3.9:** uninstall first.
2. Stop the instrumented apps (IIS w3wp processes, Windows services, standalone apps).
3. Elevated prompt: `msiexec /i dotNetAgentSetup.msi /l log.txt /q AD_CONTROLLER_ACCOUNT_ACCESS_KEY=<key>` (+ INSTALLDIR= for a custom location). The MSI keeps existing configuration.
4. Restart: **Restart IIS** in the Agent Configuration utility (or iisreset), plus services/apps.

Block 8 — Custom correlation (1.6)

When: a call goes over an **unsupported protocol or framework**, so the downstream tier appears as a separate BT.

- **Producer (caller):** custom **exit point** injects the correlation header (default key **singularityheader**) into the payload (header map, message properties).
- **Consumer (receiver): POJO entry point** reads the header and continues the **same BT**.
- Configured in **custom-activity-correlation.xml** next to app-agent-config.xml (template in the agent conf folder). Java Agent **3.8+**. Change the key with <header-key> inside <payload-operation>.
- **Not supported:** cross-process correlation by an ID stored in a file/database; **Sun RMI** (no place to carry the header).

Lab — SaaS trial (~60 min)

1. **Java Agent:** Tomcat + Spring PetClinic on Linux; application PetClinic, tier Web, node web01, SaaS host on 443 with SSL. Confirm the node and at least 3 BTs.
2. **Precedence test:** keep node-name=web01 in the XML, add -Dappdynamics.agent.nodeName=web-sysprop and export APPDYNAMICS_AGENT_NODE_NAME=web-env. Restart. Which name appears, and why?
3. **.NET Agent:** install the MSI on an IIS sample site, configure with the utility, iisreset, confirm a second tier.
4. **Break it:** set a wrong access key, restart, find the error in agent.log, fix it.
5. **Paper design:** an order service sends messages to a billing service over a proprietary TCP library and they show as separate BTs. Describe the custom correlation on each side.

Day 6 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. Node name set in controller-info.xml, as -Dappdynamics.agent.nodeName and as APPDYNAMICS_AGENT_NODE_NAME (all different). Which wins?

A. controller-info.xml
B. The system property
C. The environment variable
D. Random
Answer: C — Env vars > system properties > agent.properties > controller-info.xml; first non-empty value wins.

Q2. Which controller-info.xml element is required for every Controller?

A. account-name
B. account-access-key
C. controller-ssl-enabled
D. node-name
Answer: B — Access key is always required; account-name is required for SaaS / multi-tenant.

Q3. How do you load the Java Agent into Tomcat on Linux?

A. Copy the jar to Tomcat lib
B. Add -javaagent to CATALINA_OPTS in setenv.sh, restart
C. Deploy as a WAR
D. Register in the Controller UI
Answer: B — setenv.sh CATALINA_OPTS, then restart the JVM.

Q4. Default Java Agent log location?

A. <controller_home>/logs
B. <agent_home>/ver<version>/logs/<node_name>/
C. /var/log/appdynamics
D. Tomcat logs only
Answer: B — Start with agent.log.

Q5. Why rename the new agent folder to the old name during upgrade?

A. Controller rejects new folders
B. So the -javaagent path stays the same
C. To keep the license
D. Required for SSL
Answer: B — Startup scripts don't change; still copy config, merge changes, copy cacerts.jks, restart.

Q6. .NET config.xml default location?

A. C:\Program Files\AppDynamics
B. %ProgramData%\AppDynamics\DotNetAgent\Config\config.xml
C. web.config
D. %TEMP%
Answer: B — Edit as admin; restart coordinator then apps.

Q7. After editing config.xml for a Windows service, restart what, in order?

A. Only the Controller
B. AppDynamics.Agent.Coordinator, then the Windows service
C. The whole server
D. Nothing
Answer: B — Coordinator first, then the instrumented app.

Q8. How do you instrument a Windows service with the .NET Agent?

A. The utility detects it
B. standalone-application (executable + tier) under standalone-applications in config.xml
C. Separate MSI
D. Not supported
Answer: B — The utility only handles IIS; services and standalone apps are manual.

Q9. Upgrading a .NET Agent from 3.7 — what differs from 4.x?

A. Nothing
B. Before 3.9 you must uninstall first
C. Upgrade Events Service first
D. Reinstall IIS
Answer: B — 3.9+ upgrades in place; older versions must be uninstalled first.

Q10. Java custom correlation — what does each side do?

A. Consumer injects, producer reads
B. Producer exit point injects singularityheader; consumer POJO entry point reads it
C. Controller matches by timestamp
D. Shared database ID
Answer: B — custom-activity-correlation.xml, Java Agent 3.8+; ID-in-database correlation and Sun RMI unsupported.

Quick-review cheat sheet

Remember	Value
Precedence	Env vars > system props > agent.properties (ver, global) > controller-info.xml (ver, global)
Required always	account-access-key (account-name for SaaS / multi-tenant)
Ports	On-prem 8090 / 8181 · SaaS 443 (or 80)
Load Java agent	-javaagent:<agent_home>/javaagent.jar · Tomcat setenv.sh CATALINA_OPTS · restart JVM
Node names	Unique per application + host · reuse.nodeName for containers
Java logs	<agent_home>/ver<version>/logs/<node>/agent.log
Java upgrade	Controller first · rename folders (same path) · copy controller-info.xml, merge config, cacerts.jks · restart
.NET install	dotNetAgentSetup64.msi (elevated) · Agent Configuration utility (IIS only) · iisreset
.NET files	%ProgramData%\AppDynamics\DotNetAgent\Config\config.xml · ...\Logs
.NET coordinator	AppDynamics.Agent.Coordinator — restart it, then the apps
Windows services	standalone-applications / standalone-application executable + tier in config.xml
.NET upgrade	≥ 3.9 in place · < 3.9 uninstall first · stop w3wp · restart IIS
Custom correlation	Unsupported protocol/framework · exit point injects, POJO entry point reads · custom-activity-correlation.xml

Sources

- Configuration Options and Precedence (on-prem 26.9.0)
- Configure the Java Agent
- Use Environment Variables for Java Agent Settings
- Agent Identification Properties
- Apache Tomcat Startup Settings
- Locate the Java Agent Log Files
- Upgrade the Java Agent
- Unattended Installation for .NET
- Configure .NET Agent Properties
- Configure the .NET Agent for Windows Services and Standalone Applications
- Upgrade the .NET Agent from >= 3.9
- Custom Correlation for Java Applications (Knowledge Base)

Lab — model answers

What a correct result looks like for each Day 6 lab task, with the rule behind it.

1. Java Agent on Tomcat + PetClinic (SaaS)

```
<controller-host>YOUR-ACCOUNT.saas.appdynamics.com</controller-host>
<controller-port>443</controller-port>
<controller-ssl-enabled>true</controller-ssl-enabled>
<application-name>PetClinic</application-name>
<tier-name>Web</tier-name>
<node-name>web01</node-name>
<account-name>YOUR-ACCOUNT</account-name>
<account-access-key>YOUR-ACCESS-KEY</account-access-key>
```

- setenv.sh: export CATALINA_OPTS="$CATALINA_OPTS -javaagent:/opt/appdynamics/javaagent/javaagent.jar" → restart Tomcat.
- Click around PetClinic to generate traffic; within a few minutes the node web01 appears under Tiers & Nodes and BTs (e.g. owners, vets pages) appear.
- The account name and access key come from your SaaS Controller's license/account settings or welcome email.

2. Precedence test

Sources set	Node name shown	Why
XML + -D + env var	**web-env**	Environment variables have the highest priority
XML + -D (env var removed)	**web-sysprop**	System properties beat the files
XML only	**web01**	controller-info.xml is used only when nothing higher is set

Each test creates a new node in the Controller, since node names differ. Exam takeaway: when a node shows up with an unexpected name, look for an environment variable or -D property first.

3. .NET Agent on IIS

1. Run dotNetAgentSetup64.msi from an elevated prompt (or double-click as administrator).
2. AppDynamics Agent Configuration utility: Controller host, port 443, SSL on, account name, access key, application name.
3. Select the IIS site(s) to instrument and assign a tier (e.g. Payments-IIS).
4. Finish, then Restart IIS in the utility (or iisreset). Confirm the AppDynamics.Agent.Coordinator service is running.
5. Browse the site to generate traffic → a second tier appears on the flow map and under Tiers & Nodes.

4. Wrong access key

- The node never appears. agent.log in <agent_home>/ver<version>/logs/web01/ shows the agent failing to register / authenticate with the Controller.
- Fix the key where it is **actually** being read from — if an environment variable or -D sets it, editing controller-info.xml does nothing (precedence).
- Restart the JVM and confirm the node registers.

5. Custom correlation design: order service → billing service over a proprietary TCP library

Side	Configuration
Producer (order service)	Define a **custom exit point** on the library's send method (e.g. TcpClient.send(Message)). Add a payload operation that writes the correlation header (**singularityheader**) into the message's header map / properties.
Consumer (billing service)	Define a **POJO entry point** on the receive/handle method (e.g. BillingHandler.onMessage(Message)) and configure it to read the same header key from the payload.
Where	**custom-activity-correlation.xml** next to app-agent-config.xml on both agents (Java Agent 3.8+). Optional validation methods on each side. Restart both JVMs.
Verify	One BT now flows Order → Billing on the flow map instead of two separate BTs.
Limit	If the message has nowhere to carry the header (like Sun RMI), or the link is only an ID in a database/file, custom correlation can't join them.